

# Complex Vectors and Matrices

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## Complex vectors and matrices

**Definition 0.1** (Complex vectors and matrices). A **complex vector** of length  $p$  is a column vector whose entries are [complex numbers](#); the set of all of them is denoted  $\mathbb{C}^p$ . A **complex matrix** of size  $m \times n$  is a matrix with complex entries; the set of all of them is denoted  $\mathbb{C}^{m \times n}$ .

The **complex conjugate** of a complex vector or matrix takes the [complex conjugate](#) of each entry:  $(\overline{\mathbf{A}})_{jk} \stackrel{\text{def}}{=} \overline{a_{jk}}$ .

**Example 0.1** (A complex vector and its conjugate).

$$\underset{3 \times 1}{\tilde{z}} = \begin{bmatrix} 1+i \\ 2 \\ -3i \end{bmatrix} \in \mathbb{C}^3, \quad \underset{3 \times 1}{\tilde{\tilde{z}}} = \begin{bmatrix} 1-i \\ 2 \\ 3i \end{bmatrix}.$$

The real entry 2 is its own conjugate. Every real vector is also a complex vector, since each real number  $a$  is the complex number  $a + 0i$ , so  $\mathbb{R}^p$  is a subset of  $\mathbb{C}^p$ .

**Definition 0.2** (Conjugate transpose (Hermitian adjoint, Hermitian conjugate, transjugate)). The **conjugate transpose** (also called the **Hermitian adjoint**, **Hermitian conjugate**, or **transjugate**) of an  $m \times n$  complex matrix  $\mathbf{A}$  is the  $n \times m$  matrix  $\mathbf{A}^H$  obtained by transposing  $\mathbf{A}$  and then taking the complex conjugate of each entry:

$$(\mathbf{A}^H)_{jk} \stackrel{\text{def}}{=} \overline{a_{kj}} \quad \text{for } j \in \{1, \dots, n\}, \quad k \in \{1, \dots, m\}.$$

The conjugate transpose of a column vector  $\tilde{z} \in \mathbb{C}^p$  is the  $1 \times p$  row vector  $\tilde{z}^H = [\overline{z_1}, \dots, \overline{z_p}]$ .

*Remark 0.1* (Notation for the conjugate transpose). The operation is denoted variously across mathematics and physics:  $\mathbf{A}^*$  in pure mathematics (Axler (2024, Definition 7.7, p. 231)),  $\mathbf{A}^\dagger$  (“*A*-dagger”) in physics and quantum mechanics, and  $\mathbf{A}^H$  in engineering, numerical linear algebra, and statistics. These notes write  $\mathbf{A}^H$  because  $*$  is reserved for optimal points (as in  $\tilde{x}^*$ ) and  $^\dagger$  frequently denotes the Moore-Penrose pseudoinverse in the broader literature.

The Hermitian adjoint matrix should not be confused with the classical *adjugate matrix* (the transpose of the matrix of cofactors), which older texts occasionally referred to as the “adjoint.”

**Example 0.2** (The conjugate transpose of a  $2 \times 3$  matrix). For

$$\underbrace{\mathbf{A}}_{2 \times 3} = \begin{bmatrix} 1+i & 2 & 0 \\ -i & 3-2i & 4 \end{bmatrix},$$

transposing gives a  $3 \times 2$  matrix, and conjugating each of its entries gives

$$\underbrace{\mathbf{A}^H}_{3 \times 2} = \begin{bmatrix} 1-i & i \\ 2 & 3+2i \\ 0 & 4 \end{bmatrix}.$$

For instance,

$$\begin{aligned} (\mathbf{A}^H)_{12} &= \overline{a_{21}} \\ &= \overline{-i} \\ &= i. \end{aligned}$$

The transpose alone would have  $-i$  in that position, so for a matrix with an entry that is not real,  $\mathbf{A}^H$  and  $\mathbf{A}^\top$  differ. For a real matrix, conjugating changes nothing, so  $\mathbf{A}^H = \mathbf{A}^\top$ .

**Theorem 0.1** (Properties of the Hermitian adjoint). *Let  $\mathbf{A}$  and  $\mathbf{B}$  be complex matrices of compatible dimensions, and let  $c \in \mathbb{C}$  be a scalar. The Hermitian adjoint (Definition 0.2) satisfies:*

1. ***Involution:***

$$(\mathbf{A}^H)^H = \mathbf{A}$$

2. ***Additivity:***

$$(\mathbf{A} + \mathbf{B})^H = \mathbf{A}^H + \mathbf{B}^H$$

3. ***Conjugate-homogeneity:***

$$(c\mathbf{A})^H = \bar{c}\mathbf{A}^H$$

4. ***Reversal of products:***

$$(\mathbf{AB})^H = \mathbf{B}^H \mathbf{A}^H$$

If  $\mathbf{A}$  is invertible, then  $\mathbf{A}^H$  is also invertible, and

$$(\mathbf{A}^H)^{-1} = (\mathbf{A}^{-1})^H.$$

### **i** Proof

*Proof.* Properties 1-3 follow directly from entrywise conjugation and transposition:

- For involution:

$$\begin{aligned} [(\mathbf{A}^H)^H]_{jk} &= \overline{(\mathbf{A}^H)_{kj}} \\ &= \overline{\overline{a_{jk}}} \\ &= a_{jk}. \end{aligned}$$

- For additivity:

$$\begin{aligned} [(\mathbf{A} + \mathbf{B})^H]_{jk} &= \overline{(\mathbf{A} + \mathbf{B})_{kj}} \\ &= \overline{a_{kj} + b_{kj}} \\ &= \overline{a_{kj}} + \overline{b_{kj}} \\ &= (\mathbf{A}^H)_{jk} + (\mathbf{B}^H)_{jk}. \end{aligned}$$

- For conjugate-homogeneity:

$$\begin{aligned} [(c\mathbf{A})^H]_{jk} &= \overline{(c\mathbf{A})_{kj}} \\ &= \overline{c a_{kj}} \\ &= \overline{c} \overline{a_{kj}} \\ &= \overline{c} (\mathbf{A}^H)_{jk}. \end{aligned}$$

Unlike the real transpose where  $(c\mathbf{A})^\top = c\mathbf{A}^\top$ , scaling conjugates the scalar.

For property 4 (reversal of products), let  $\mathbf{A}$  be an  $m \times k$  matrix and  $\mathbf{B}$  a  $k \times n$  matrix. Entry  $(i, j)$  of the product's Hermitian adjoint (for  $i \in \{1, \dots, n\}$  and  $j \in \{1, \dots, m\}$ ) is:

$$\begin{aligned}
[(\mathbf{AB})^H]_{ij} &= \overline{(\mathbf{AB})_{ji}} && \text{(Definition 0.2)} \\
&= \overline{\sum_{s=1}^k a_{js} b_{si}} && \text{(definition of matrix multiplication)} \\
&= \sum_{s=1}^k \overline{a_{js}} \overline{b_{si}} && \text{(conjugate of sums and products)} \\
&= \sum_{s=1}^k \overline{b_{si}} \overline{a_{js}} && \text{(complex multiplication is commutative)} \\
&= \sum_{s=1}^k (\mathbf{B}^H)_{is} (\mathbf{A}^H)_{sj} && \text{(Definition 0.2)} \\
&= (\mathbf{B}^H \mathbf{A}^H)_{ij} && \text{(definition of matrix multiplication).}
\end{aligned}$$

For the inverse, applying property 4 to  $\mathbf{AA}^{-1} = \mathbf{I}$  gives:

$$\begin{aligned}
(\mathbf{A}^{-1})^H \mathbf{A}^H &= (\mathbf{AA}^{-1})^H \\
&= \mathbf{I}^H \\
&= \mathbf{I},
\end{aligned}$$

so  $(\mathbf{A}^H)^{-1} = (\mathbf{A}^{-1})^H$ . □

**Theorem 0.2** (The adjoint property for inner products). *Let  $\mathbf{A}$  be an  $m \times n$  complex matrix. Under the standard inner product on complex Euclidean space,  $\langle \tilde{u}, \tilde{v} \rangle \stackrel{\text{def}}{=} \tilde{u}^H \tilde{v}$ ,*

$$\langle \mathbf{A}\tilde{u}, \tilde{v} \rangle = \langle \tilde{u}, \mathbf{A}^H \tilde{v} \rangle \quad \text{for all } \tilde{u} \in \mathbb{C}^n \text{ and } \tilde{v} \in \mathbb{C}^m.$$

### **i** Proof

*Proof.* By the definition of the standard complex inner product and the product rule for the Hermitian adjoint (Theorem 0.1):

$$\begin{aligned}
\langle \mathbf{A}\tilde{u}, \tilde{v} \rangle &= (\mathbf{A}\tilde{u})^H \tilde{v} && \text{(definition of standard inner product)} \\
&= (\tilde{u}^H \mathbf{A}^H) \tilde{v} && \text{(Theorem 0.1, reversal of products)} \\
&= \tilde{u}^H (\mathbf{A}^H \tilde{v}) && \text{(associativity of matrix multiplication)} \\
&= \langle \tilde{u}, \mathbf{A}^H \tilde{v} \rangle && \text{(definition of standard inner product).}
\end{aligned}$$
□

*Remark 0.2* (Where the name “adjoint” comes from). This identity is the origin of the name **adjoint**: in functional analysis and operator theory, the *adjoint* of a linear map

$T : V \rightarrow W$  between inner product spaces is defined as the unique operator  $T^* : W \rightarrow V$  satisfying  $\langle T\tilde{u}, \tilde{v} \rangle = \langle \tilde{u}, T^*\tilde{v} \rangle$  (Axler (2024, Definition 7.1, p. 226)). When  $V$  and  $W$  are finite-dimensional spaces equipped with their standard orthonormal bases, the matrix representing the adjoint operator  $T^*$  is precisely the Hermitian adjoint matrix  $\mathbf{A}^H$  (Axler (2024, Proposition 7.9, p. 231)).

**Theorem 0.3** ( $\tilde{z}^H \tilde{z}$  is a sum of squared absolute values). *For every  $\tilde{z} \in \mathbb{C}^p$ ,*

$$\underbrace{\tilde{z}^H}_{1 \times p} \underbrace{\tilde{z}}_{p \times 1} = \sum_{j=1}^p |z_j|^2,$$

*which is a nonnegative real number, and is 0 only when  $\tilde{z} = \tilde{0}_{p \times 1}$ .*

### **i** Proof

*Proof.* The last step uses the [identity for a complex number times its conjugate](#).

$$\begin{aligned} \tilde{z}^H \tilde{z} &= \sum_{j=1}^p \overline{z_j} z_j \quad (\text{Definition 0.2, and the row-times-column product}) \\ &= \sum_{j=1}^p z_j \overline{z_j} \quad (\text{complex multiplication is commutative}) \\ &= \sum_{j=1}^p |z_j|^2 \quad (\text{a number times its conjugate}) \end{aligned}$$

By the definition of the [absolute value](#), each  $|z_j|^2 = (\operatorname{Re} z_j)^2 + (\operatorname{Im} z_j)^2$  is a nonnegative real number, so their sum is too. A sum of nonnegative numbers is 0 only when every term is 0, and  $|z_j|^2 = 0$  only when

$$\begin{aligned} \operatorname{Re} z_j &= \operatorname{Im} z_j \\ &= 0, \end{aligned}$$

that is, when  $z_j = 0$ . □

**Example 0.3** (Computing  $\tilde{z}^H \tilde{z}$ ). Let  $\tilde{z} = (1 + i, 2)^\top$ . By Theorem 0.3,

$$\begin{aligned}
\widetilde{\widetilde{\mathbf{z}}}^{\mathbf{H}}_{1 \times 2} \widetilde{\widetilde{\mathbf{z}}}_{2 \times 1} &= |1+i|^2 + |2|^2 && \text{(Theorem 0.3)} \\
&= (1^2 + 1^2) + (2^2 + 0^2) && \text{(definition of the absolute value, squared)} \\
&= 2 + 4 && \text{(add inside each group)} \\
&= 6 && \text{(add)}
\end{aligned}$$

Multiplying out directly gives the same number:  $\overline{2} \cdot 2 = 4$ , and

$$\begin{aligned}
\overline{(1+i)}(1+i) &= (1-i)(1+i) && \text{(conjugate)} \\
&= 1+i-i-i^2 && \text{(distribute)} \\
&= 1-i^2 && \text{(cancel } i) \\
&= 2 && (i^2 = -1)
\end{aligned}$$

**Example 0.4** (Why complex vectors use the conjugate transpose). For a real vector  $\tilde{x}$ ,  $\tilde{x}^\top \tilde{x} = \sum_{j=1}^p x_j^2$  is 0 only when  $\tilde{x}$  is the zero vector. The same formula fails for complex vectors. Let  $\tilde{z} = (1, i)^\top$ . Then

$$\begin{aligned}
\widetilde{\widetilde{\mathbf{z}}}^\top_{1 \times 2} \widetilde{\widetilde{\mathbf{z}}}_{2 \times 1} &= 1 \cdot 1 + i \cdot i && \text{(multiply entry by entry and add)} \\
&= 1 + i^2 && \text{(multiply)} \\
&= 0 && (i^2 = -1)
\end{aligned}$$

although  $\tilde{z} \neq \tilde{0}_{2 \times 1}$ . Using the conjugate transpose instead,

$$\begin{aligned}
\widetilde{\widetilde{\mathbf{z}}}^{\mathbf{H}}_{1 \times 2} \widetilde{\widetilde{\mathbf{z}}}_{2 \times 1} &= \overline{1} \cdot 1 + \overline{i} \cdot i && \text{(Definition 0.2)} \\
&= 1 + (-i) \cdot i && \text{(conjugate each entry)} \\
&= 1 - i^2 && \text{(multiply)} \\
&= 2 && (i^2 = -1)
\end{aligned}$$

which is positive, as Theorem 0.3 guarantees.

**Definition 0.3** (Hermitian matrix (self-adjoint matrix)). A square complex matrix  $\mathbf{A} \in \mathbb{C}^{n \times n}$  is **Hermitian** (or **self-adjoint**) if it equals its conjugate transpose (Definition 0.2):

$$\mathbf{A}^{\mathbf{H}} = \mathbf{A}.$$

*Remark 0.3* (Why “self-adjoint”). The term *self-adjoint* comes directly from Theorem 0.2: when  $\mathbf{A}$  is Hermitian,  $\langle \mathbf{A}\tilde{u}, \tilde{v} \rangle = \langle \tilde{u}, \mathbf{A}\tilde{v} \rangle$  for all  $\tilde{u}, \tilde{v} \in \mathbb{C}^n$ . The linear operator moves across the inner product without change.

**Example 0.5** (A Hermitian matrix).

$$\underset{2 \times 2}{\mathbf{A}} = \begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix}$$

is Hermitian: transposing swaps the off-diagonal entries, and conjugating turns  $1+i$  back into  $1-i$  and  $1-i$  back into  $1+i$ , so  $\mathbf{A}^H = \mathbf{A}$ . The diagonal entries of a Hermitian matrix must be real, because  $a_{jj} = \overline{a_{jj}}$ , and only a real number is its own conjugate: if  $a + bi = a - bi$ , then the imaginary parts give  $b = -b$ , so  $b = 0$ .

Every real symmetric matrix is Hermitian, because for a real matrix  $\mathbf{A}^H = \mathbf{A}^\top$ .

**Example 0.6** (A symmetric complex matrix that is not Hermitian).

$$\underset{2 \times 2}{\mathbf{B}} = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$$

is symmetric,  $\mathbf{B}^\top = \mathbf{B}$ , but it is not Hermitian:

$$\underset{2 \times 2}{\mathbf{B}}^H = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \neq \mathbf{B}.$$

**Definition 0.4** (Skew-Hermitian matrix (anti-Hermitian matrix)). A square complex matrix  $\mathbf{A} \in \mathbb{C}^{n \times n}$  is **skew-Hermitian** (or **anti-Hermitian**) if it equals the negative of its conjugate transpose (Definition 0.2):

$$\mathbf{A}^H = -\mathbf{A}.$$

*Remark 0.4* (The diagonal of a skew-Hermitian matrix). Every entry of a skew-Hermitian matrix satisfies  $a_{jk} = -\overline{a_{kj}}$ . In particular, its diagonal entries satisfy  $a_{jj} = -\overline{a_{jj}}$ , which means

$$\begin{aligned} a_{jj} + \overline{a_{jj}} &= 2 \operatorname{Re}(a_{jj}) \\ &= 0. \end{aligned}$$

Therefore, every diagonal entry of a skew-Hermitian matrix must be purely imaginary (or zero). For example,  $a_{jj} = 3i$  works, since

$$\begin{aligned} -\overline{3i} &= -(-3i) \\ &= 3i, \end{aligned}$$

but  $a_{jj} = 3$  does not, since  $-\overline{3} = -3$ .

**Example 0.7** (A skew-Hermitian matrix). The matrix

$$\mathbf{A} = \begin{bmatrix} 2i & 1+i \\ -1+i & 0 \end{bmatrix}$$

is skew-Hermitian: transposing and conjugating gives

$$\begin{aligned} \mathbf{A}^H &= \begin{bmatrix} \overline{2i} & \overline{-1+i} \\ \overline{-1+i} & \overline{0} \end{bmatrix} \\ &= \begin{bmatrix} -2i & -1-i \\ 1-i & 0 \end{bmatrix} \\ &= -\mathbf{A}. \end{aligned}$$

Its diagonal entries  $2i$  and  $0$  are purely imaginary.

**Theorem 0.4** (Hermitian and skew-Hermitian decomposition). *Every square complex matrix  $\mathbf{A} \in \mathbb{C}^{n \times n}$  can be uniquely expressed as the sum of a Hermitian matrix  $\mathbf{H}$  (Definition 0.3) and a skew-Hermitian matrix  $\mathbf{S}$  (Definition 0.4):*

$$\mathbf{A} = \mathbf{H} + \mathbf{S},$$

where

$$\mathbf{H} = \frac{\mathbf{A} + \mathbf{A}^H}{2} \quad \text{and} \quad \mathbf{S} = \frac{\mathbf{A} - \mathbf{A}^H}{2}.$$

### **i** Proof

*Proof.* **Existence:** Define  $\mathbf{H} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^H)$  and  $\mathbf{S} = \frac{1}{2}(\mathbf{A} - \mathbf{A}^H)$ . Their sum is:

$$\begin{aligned} \mathbf{H} + \mathbf{S} &= \frac{\mathbf{A} + \mathbf{A}^H}{2} + \frac{\mathbf{A} - \mathbf{A}^H}{2} \\ &= \frac{2\mathbf{A}}{2} \\ &= \mathbf{A}. \end{aligned}$$

Applying the properties of the Hermitian adjoint (Theorem 0.1):

$$\begin{aligned}
\mathbf{H}^H &= \left( \frac{\mathbf{A} + \mathbf{A}^H}{2} \right)^H \\
&= \frac{\mathbf{A}^H + (\mathbf{A}^H)^H}{2} \\
&= \frac{\mathbf{A}^H + \mathbf{A}}{2} \\
&= \mathbf{H}, \\
\mathbf{S}^H &= \left( \frac{\mathbf{A} - \mathbf{A}^H}{2} \right)^H \\
&= \frac{\mathbf{A}^H - (\mathbf{A}^H)^H}{2} \\
&= \frac{\mathbf{A}^H - \mathbf{A}}{2} \\
&= -\mathbf{S}.
\end{aligned}$$

Thus  $\mathbf{H}$  is Hermitian and  $\mathbf{S}$  is skew-Hermitian.

**Uniqueness:** Suppose  $\mathbf{A} = \mathbf{H}' + \mathbf{S}'$  with  $(\mathbf{H}')^H = \mathbf{H}'$  and  $(\mathbf{S}')^H = -\mathbf{S}'$ . Taking the Hermitian adjoint of both sides yields:

$$\begin{aligned}
\mathbf{A}^H &= (\mathbf{H}' + \mathbf{S}')^H \\
&= (\mathbf{H}')^H + (\mathbf{S}')^H \\
&= \mathbf{H}' - \mathbf{S}'.
\end{aligned}$$

Adding  $\mathbf{A} = \mathbf{H}' + \mathbf{S}'$  and  $\mathbf{A}^H = \mathbf{H}' - \mathbf{S}'$  gives  $\mathbf{A} + \mathbf{A}^H = 2\mathbf{H}'$ , so

$$\begin{aligned}
\mathbf{H}' &= \frac{1}{2}(\mathbf{A} + \mathbf{A}^H) \\
&= \mathbf{H}.
\end{aligned}$$

Subtracting the two equations gives  $\mathbf{A} - \mathbf{A}^H = 2\mathbf{S}'$ , so

$$\begin{aligned}
\mathbf{S}' &= \frac{1}{2}(\mathbf{A} - \mathbf{A}^H) \\
&= \mathbf{S}.
\end{aligned}$$

□